

Retaining Rings

External C-Rings / E-Rings

Retaining Rings – External C-Ring

RoHS 10

STWN
STWS Stainless Steel

No.8 or Less

Applicable Shaft

d₄: Maximum outer diameter when retaining ring is fitted on d₁ (Clearance outer diameter)

Type	Material	Hardness
STWN	Spring Steel	44–53 HRC min.
STWS	304 Stainless Steel-CSP	36–44 HRC min.

Part Number		Retaining Ring							Reference	Applicable Shaft (Reference)						
Type	No.	d ₃	Tolerance	t	Tolerance	b (Approx.)	a (Approx.)	d ₀ (Min.)	d ₄	d ₁	d ₂	Tolerance	m	Tolerance	n (Min.)	
STWN	3	2.7	+0.04 -0.15	0.25	±0.025	0.5	1.7	0.7	7	3	2.85	0 -0.04	0.35	+0.10 0	0.3	
	4	3.7		0.4	±0.03	0.9	2.2	0.8	9	4	3.8		0.5			
	5	4.7	0.6	±0.04	1.1	2.4	1	10.5	5	4.8	0.7					
	6	5.6	0.7		1.3	2.8		12	6	5.7	0.8					
	7	6.5	1.4		3	14		7	6.7	0.9						
8	7.4	0.8	1.6	3	15	8		7.6	-0.06	0.9	0.6					
STWS Stainless Steel	4	3.6	±0.1	0.25	±0.03	0.7		1.4		0.7	9	4	3.7	0.3	+0.10 0	1.0
	5	4.5		0.4	0.8		2.0	10.5			5	4.7	0.45			
	6	5.5		0.65	0.9		1	12	6		5.7	±0.06	0.7			
	7	6.3			0.9	7		6.5	0.7							
8	7.1	1.0		2.2	15	8	7.4									
STWN STWS Stainless Steel	10	9.3		±0.15	1	±0.05	1.8	3	1.2	17	10	9.6	0 -0.09	1.15	+0.14 0	1.5
	11	10.2					2	3.1		18	11	10.5				
	12	11.1					2.1	3.2		1.5	19	12	11.5			
	13	12	3.3					20			13	12.4				
	14	12.9	2.2	3.4				1.7	22		14	13.4				
	15	13.8		3.5					23		15	14.3				
	16	14.7		3.6			24		16	15.2						
	17	15.7		3.7			25		17	16.2						
	18	16.5	1.2	2.6	3.8	2	26	18	17	0 -0.21	1.35					
	19	17.5		2.7	3.9		27	19	18							
	20	18.5			4		28	20	19							
	21	19.5			4.1		30	21	20							
	22	20.5			2.9		4.5	31	22		21	1.65				
	23	21.4		3.1	4.2		32.5	23	22							
	24	22.2			4.3		33	24	22.9							
	25	23.2			4.4		34	25	23.9							
	26	24.2	1.5		±0.06	3.5	4.6	35	26	24.9	0 -0.25	1.9				
	28	25.9		4.7			38	28	26.6							
	29	26.9		4.8			39	29	27.6							
	30	27.9		5			40	30	28.6							
	32	29.6	±0.25	1.75		±0.07	4	5.4	43	32		30.3	0 -0.3	2.7		
	35	32.2					4.5	5.8	46	35		33				
	40	37					4.8	6.3	53	40		38				
	45	41.5					5	6.7	58	45		42.5				
	50	45.8	±0.4	2	±0.08	6.8	64	50	47	2.2	2					
	52	47.8				7.2	66	52	49							
	60	55.8				7.4	8.2	75	60			57				
	80	74.5				7.4	8.2	97	80			76.5				

Cost-saving box package is available when large quantity is required.

Retaining Rings – E-Ring

RoHS 10

NETW
NETWS Stainless Steel

Applicable Shaft

Type	Material	Hardness	Surface Treatment
NETW	Spring Steel	44–53 HRC min.	Trivalent Chromate Plating
NETWS	304 Stainless Steel-CSP	37–46 HRC min.	—

Part Number		Retaining Ring								Applicable Shaft (Reference)							
Type	No.	d	Tolerance	D	Tolerance	H	Tolerance	t	Tolerance	d ₁ Class		d ₂	Tolerance	m	Tolerance	n (Min.)	
NETW	1.2	1.2	0 -0.09	3	±0.1	1	0 -0.25	0.3	±0.025	More than	or Less	1.2	+0.06 0	0.4	+0.05 0	0.6	
	1.5	1.5		4		1.3		2		2.5	1.5	0.8					
	2	2		5		1.7		2.5		3.2	2	1					
	2.5	2.5		6		2.1		3.2		4	2.5						
	3	3		7		2.6		4		5	3						
NETWS Stainless Steel	4	4	0 -0.12	9	±0.2	3.5	0 -0.30	0.6	±0.04	5	7	4	+0.075 0	0.7	+0.1 0	1.2	
	5	5		11		4.3		6		8	5						
	6	6		12		5.2		7		9	6						
	7	7		14		6.1		0.8		8	11	7		+0.09 0		0.9	1.5
	8	8		16		6.9				9	12	8					1.8
	9	9		18		7.8				10	14	9					2

Part Number Example

STWS6
NETW9

Part Number

Check out misumiusa.com for the most current pricing and lead time.